

# Ribbon Links: Classification up to 11 Crossings

## Group N: Final Report

Weibing Guo<sup>1</sup>   Chloe Comeau<sup>2</sup>   Yeteng Tang<sup>3</sup>   Cahit Erdem  
Anil<sup>4</sup>   Giovanni Picone<sup>5</sup>

<sup>1</sup>University of Illinois Urbana-Champaign

<sup>2</sup>Vassar College

<sup>3</sup>University of Cambridge

<sup>4</sup>Boğaziçi University

<sup>5</sup>St. John's University

2025

# Background Information

## Knot & Link

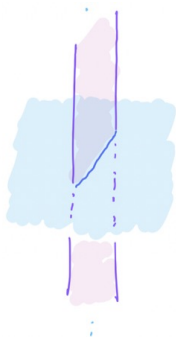
A **knot** is an embedding of  $S^1$  into  $\mathbb{R}^3$ .

A **link** is a collection of knots that do not intersect, but which may be linked (or knotted) together.

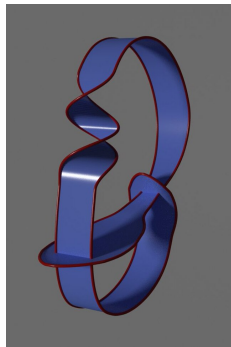
## Ribbon Knot

A **ribbon knot** is a knot that bounds a self-intersecting disk with only **ribbon singularities**.

*Goal: Classify links up to 11 crossings as ribbon or non-ribbon.*



Ribbon singularity



Ribbon presentation

## Checking if links from LinkInfo are ribbon links

Attempting to prove one of the following open questions:

- *Does every ribbon knot  $K$  admit a symmetric union presentation?*
- *Does every ribbon link admit a symmetric union presentation?*

# Criteria for Ribbon Links

For a link to be ribbon, it must satisfy specific invariants. We filtered the *LinkInfo* database using these criteria:

- ① **Smooth Four-Genus:** Must be 0. (Every ribbon knot is slice).
- ② **Linking Matrix:** All entries must be 0.
- ③ **Jones Polynomial:** Must be divisible by  $(q + q^{-1})^{n-1}$  for an  $n$ -component link.
- ④ **Determinant:** Must satisfy modular equivalence conditions.

# The Candidate List

Initial filtering on *LinkInfo* (Genus=0, Linking Matrix=0) gave us a preliminary list. However, several links had **undetermined** 4D-genus (e.g.,  $[0, 1]$ ). We included these to be rigorous.

## Potential Candidates (13 Links)

- |                    |                    |
|--------------------|--------------------|
| • L9n27 (3-comp)   | • L11n244 (3-comp) |
| • L10n32 (2-comp)  | • L11n396 (3-comp) |
| • L10n36 (2-comp)  | • L11n406 (3-comp) |
| • L10n56 (2-comp)  | • L11n426 (3-comp) |
| • L10n57 (2-comp)  | • L11n433 (3-comp) |
| • L10n59 (2-comp)  | • L11n436 (3-comp) |
| • L10n107 (4-comp) |                    |

# Jones Polynomial Divisibility Test

## Theorem (Eisermann)

*For every  $n$ -component ribbon link  $L$ , the Jones polynomial  $V(L)$  is divisible by the polynomial of the unlink  $V(\bigcirc^n) = (q + q^{-1})^{n-1}$ .*

Using SageMath & SnapPy, we tested the candidates:

- ~~L9n27~~: Fail
- L10n32: Pass
- L10n36: Pass
- L10n56: Pass
- L10n57: Pass
- L10n59: Pass
- ~~L10n107~~: Fail
- L11n244: Pass
- ~~L11n396~~: Fail
- ~~L11n406~~: Fail
- ~~L11n426~~: Fail
- ~~L11n433~~: Fail
- ~~L11n436~~: Fail

# Determinant Modular Equivalence

## Theorem

*Every  $n$ -component ribbon link  $L = K_1 \cup \cdots \cup K_n$  satisfies:*

$$\det V(L) \equiv \det(K_1) \cdots \det(K_n) \pmod{32}$$

**Survivors from previous step:** L10n32, L10n36, L10n56, L10n57, L10n59, L11n244.



# Determinant Tests: Failures

**L10n56:**

$$\det V(L) = 17, \quad \det(K_1) = 1, \quad \det(K_2) = 1$$

$$17 \not\equiv 1 \cdot 1 \pmod{32} \implies \textbf{Not Ribbon}$$

**L10n57:**

$$\det V(L) = -15, \quad \det(K_1) = 1, \quad \det(K_2) = 1$$

$$-15 \not\equiv 1 \pmod{32} \implies \textbf{Not Ribbon}$$

**L11n244:**

$$\det V(L) = 33, \quad \det(K_1) = 1, \quad \det(K_2) = -1$$

$$33 \not\equiv -1 \pmod{32} \implies \textbf{Not Ribbon}$$

# Determinant Tests: Passes

The following links pass the determinant test:

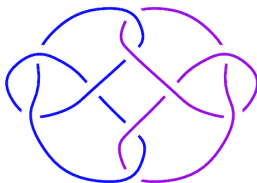
- **L10n32:**  $-23 \equiv 9 \cdot 1 \pmod{32}$ . (Pass)
- **L10n36:**  $-23 \equiv 9 \cdot 1 \pmod{32}$ . (Pass)
- **L10n59:**  $-23 \equiv (-3) \cdot (-3) \pmod{32}$ . (Pass)

**Remaining Candidates: L10n32, L10n36, L10n59**

# Analysis of L10n59

Although L10n59 passes the polynomial tests, it is a **Pretzel Link**.

- L10n59 is the pretzel link  $P(2, -3, -2, 3)$ .
- **Theorem (Aceto et al.):** Two-component pretzel links of the form  $P(p, q, -p, -q)$  are not slice.
- Since every ribbon knot is slice, **L10n59 is not ribbon**.



## Symmetric Unions: L10n32

If a link admits a **symmetric union presentation**, it is ribbon.

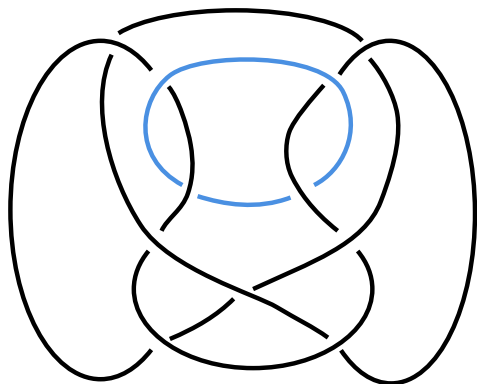


Figure: Symmetric Union of L10n32

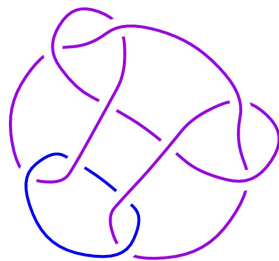


Figure: L10n36

Both L10n32 and L10n36 admit symmetric union presentations.

$\Rightarrow$  **Both are Ribbon.**

# Results Summary

| Link Name | Ribbon? |
|-----------|---------|
| L10n32    | Yes     |
| L10n36    | Yes     |
| L10n56    | No      |
| L10n57    | No      |
| L10n59    | No      |
| L9n27     | No      |
| L11n244   | No      |
| L11n396   | No      |
| L11n406   | No      |
| L11n426   | No      |
| L11n433   | No      |
| L11n436   | No      |

All other links with crossings  $\leq 11$  are not ribbon.

# Conclusion

- We systematically categorized all links with crossing number  $\leq 11$ .
- **Filters applied:**
  - 1 Smooth 4-genus = 0
  - 2 Linking Matrix = 0
  - 3 Jones Polynomial Divisibility
  - 4 Determinant Modulo 32
  - 5 Pretzel Link Slice Obstructions (Aceto et al.)
- **Result:** Only **L10n32** and **L10n36** are ribbon links.

# References



Aceto, Paolo, et al.

"Rational concordance of knots and pretzel links."

*Algebraic & Geometric Topology* 18 (2018).



Eisermann, Michael.

"The Jones polynomial of ribbon links."

*Geometry & Topology* 13.2 (2009).



Lamm, Christoph.

"Symmetric unions and ribbon knots."

*Osaka Journal of Mathematics* 37 (2000).



Livingston, C. and Moore, A.H.

*LinkInfo: Table of Knot Invariants.*

<http://linkinfo.org>